Zinal Jain

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RESEARCH INTERESTS

UAV autonomy, flight control, and edge AI for onboard perception; embedded systems and IoT.

EDUCATION

G H Patel College of Engineering & Technology, Anand, Gujarat

B.Tech in Computer Science Engineering (IoT) | Minor in Robotics
July 2023 - May 2027 (expected) | **CGPA: 8.56/10**

RESEARCH EXPERIENCE

Research Intern - DRDO, Defence Technology & Test Centre

Lucknow, India | August 2026 - Present

Edge AI-Based Human Detection and Alert System for Perimeter Surveillance

- Built and tested a laptop prototype for video capture, motion detection, multi-frame confirmation, and event data generation.
- Pipeline: motion detection - MobileNet-SSD V2 on ONNX Runtime - multi-frame confirmation - event record.
- Planned a hardware setup using Raspberry Pi, a camera, EBYTE E22 LoRa, and ESP32. Hardware implementation is proposed work.

SELECTED PROJECTS

Neerkavach - Semi-Autonomous Water Rescue Drone

2026 - ongoing | Team project | GUJCOST Robofest 6.0 Hackathon

Designed to fly to a person in distress on water and land nearby. Two thrusters move a flotation tube toward the person to provide support until help arrives. The design includes two depth cameras for landing decisions.

- **My contribution:** Flight-controller calibration and setup, waypoint mission planning, and operating semi-autonomous flights from a laptop ground station.
- **Result:** The team built a working water rescue emergency drone. Waypoint mission and water landing completed in tests.
- **Simulation:** Drowning-person detection validated in simulation.
- **In development:** Depth-camera-based landing and onboard detection by the team.

Recognition: National Finalist, GUJCOST Robofest 6.0 Hackathon
Team project funding: INR 50,000

SELECTED PROJECTS - CONTINUED

ARCNET - Aerial Robotics Minefield Navigation | 2025

Team project | National Finalist, GUJCOST Robofest 5.0
Team project grant: INR 250,000

- **Purpose:** Drone navigation for a minefield-navigation challenge.
- **My contribution:** Built the Raspberry Pi 4-Pixhawk MAVLink communication pipeline for commands and telemetry, set up and tuned ArduPilot for waypoint navigation, and worked on system integration and documentation.
- **Result:** The team produced a working drone navigation prototype.

KRAKEN - Underwater Remotely Operated Vehicle | 2024

Team project | National Finalist, GUJCOST Robofest 4.0
Team project grant: INR 250,000

- **My contribution:** Worked on project documentation and helped with wiring and assembly.
- **Result:** The team produced a working eight-thruster, six-degree-of-freedom underwater robot.

Additional recognition: Second Prize and INR 10,000 in project funding at the SSIP Competition.

TECHNICAL SKILLS

Programming: Python, C/C++
Platforms: Raspberry Pi, Pixhawk
Robotics and tools: ArduPilot, MAVLink, Mission Planner, QGroundControl

CERTIFICATIONS

SAP Code Unnati - AI/ML & Edge Computing
Ethical Hacking - SVNIT Surat